Econ 508 PS#5 Question 4

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Productivity Analysis
Sample Selection Effect

- 1. Estimate the probit model,
  $$\text{probit}(\text{P}(\text{quit} = 0) = \gamma_0 + \gamma_1 sex + \gamma_2 dex + \gamma_3 lex + \gamma_4 lex^2 + u$$

- 2. Construct the Mill's ratio, $\lambda_\text{i} = \frac{\varphi(x_i{}'\gamma)}{\Phi(x_i'\gamma)}$

- 3. Run the OLS model,
  $$\text{y} = \alpha_0 + \alpha_1 sex + \alpha_2 dex + \alpha_3 lex + \alpha_4 lex^2 + \alpha_5 \lambda_i + u$$

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	Heckman Step 1	Heckman Step 2	Full OLS	Restrictive OLS
Intercept	-3.549	14.453*	5.528**	6.994**
	(2.441)	(5.655)	(2.032)	(2.533)
sex	-0.268*	-0.516*	-0.901***	-0.901***
	(0.109	(0.282)	(0.087)	(0.106)
dex	0.053***	0.035	0.112***	0.111***
	(0.008)	(0.052)	(0.006)	(0.008)
lex	0.313	0.146	0.821*	0.600
	(0.386)	(0.502)	(0.321)	(0.398)
Lex2	-0.012	-0.010	-0.036	-0.028*
	(0.015)	(0.020)	(0.013)	(0.016)
λ	-3.227			
	(2.188)			
Obs	683	500	683	500
R^2_{adj}		0.326	0.385	0.324

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- Assumption underlying Heckman's method: error term is normally distributed.
- KS (p-Value=0.14) JB (p-value=0.45) test fail to reject the null hypothesis.
- λ is not significant, which means that there is no sample selection problem.

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- in theory, the full sample estimators are consistent and efficient, while the restrictive sample estimators are also consistent provided no sample selection bias.
- As expected, the full sample and restrictive sample estimators are pretty similar, but much differs with Heckman's.
- The difference may be caused by the high correlation between λ and.
- Moreover, the estimates from full sample are more significant than those from restrictive one, which suggests that sample selection do make effects.